# OP SILENT PERIMETER- FINAL INCIDENT RESPONSE REPORT

## 1. What happened first?

The first event was the **loss of routing to Sector Charlie (Logistics)** . This disrupted internal routing between command posts of HQ and C.

## 2. Which evidence proves credential theft?

Evidence was captured across multiple detection layers:

2.1 **Wireshark Analysis:** Packet captures revealed **plaintext usernames and passwords** transmitted during FTP and Telnet login sessions.
2.2 **Suricata IDS Logs:** Custom alerts (Rule 4) triggered when **"USER" or "PASS"** strings were observed in network traffic.

## 3. Which host was the attacker?

The primary source of the malicious activity was an **internal laptop**. Forensic investigation indicated this host was responsible for:

3.1 Launching **reconnaissance scans** against the brigade network.
3.2 Initiating an unauthorized connection.
3.3 Performing unauthorized **FTP transfers** of sensitive files from internal servers.

## 4. Which defensive control failed or was missing?

Several critical defensive layers were absent or misconfigured:

**4.1 Use of Legacy Protocols:** Management systems relied on **Telnet and FTP**, which lack encryption and expose credentials to sniffing.
**4.2 Routing Vulnerabilities:** Routing instability occurred due to a lack of robust redistribution or misconfiguration between **OSPF and EIGRP** domains.

## 5. How would you contain the incident in 10 minutes?

**5.1 Isolate the Host:** Immediately disconnect the suspicious laptop from the Sector Alpha or Bravo LAN to stop beaconing.
**5.2 Shut Down Plaintext Services:** Temporarily disable the **FTP and Legacy Telnet servers** to prevent further credential exposure.
**5.3 Block Malicious IPs/Ports:** Update router ACLs to block the suspicious foreign IP
**5.4 Stabilize Routing:** Correct network statements or wildcard masks in **R1, R2, or R3** to restore command post connectivity.

**6. Long-term remediation plan?**

- **6.1 Decommission Legacy Systems:** Retire management systems that cannot support secure communication.
- **6.2 Harden Network Infrastructure:** Implement proper route redistribution and passive interfaces to prevent routing instability.
- **6.3 Deploy Host Monitoring:** Establish regular **netstat** audits to identify unauthorized established sessions.
- **6.4 Continuous IDS Tuning:** Maintain and expand **Suricata custom rules**, including those for brute-force attempts.

**7. Which protocols should be replaced and why?**

The following protocols must be replaced to prevent credential theft and data interception:

- **7.1 Replace Telnet with SSH:** SSH provides encrypted management access, preventing plaintext password capture.
- **7.2 Replace FTP with SFTP:** SFTP ensures file transfers and authentication are fully encrypted.
- **7.3 Replace HTTP with HTTPS:** Encrypting the **DVWA web server** login prevents the capture of HTTP POST credentials.

**8. How to improve monitoring posture?**

- **8.1 Centralized IDS Alerting:** Monitor **fast.log and alerts.json** screenshots in real-time to prioritize high-risk alerts like credential exposure.
- **8.2 Threshold-Based Detection:** Implement rules to detect **multiple port hits** (e.g., 5 ports in 10 seconds) to catch scanners early.
- **8.3 Network Visibility:** Use **Wireshark** for periodic deep packet inspection of sensitive transit links between Sector Alpha, Bravo, and Charlie.